

CHAPTER 3

THE REAL-WORLD LINE

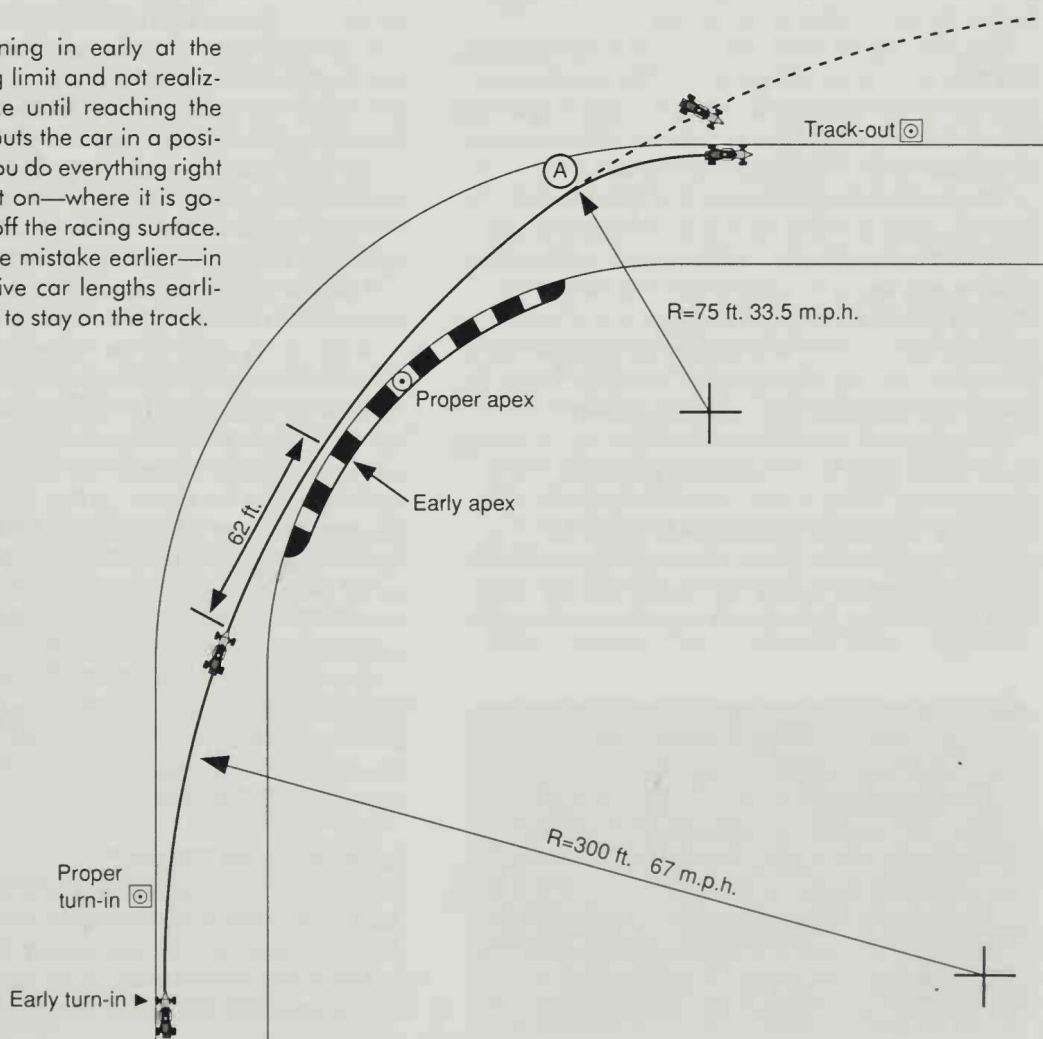
WHEN WE LOOKED AT THE basics of racing we outlined the fundamentals for driving on the best path around the racecourse. But in the real world of racing, the line is more complex. There are a number of variables you have to consider as you plan your way through the real-world line.

Real-world situations go beyond the basics of the line. There are many variables that will affect your choice of the best path through a given corner.

SPEED CHANGES THINGS

Even with the simplest corners the primary symptom of an early apex changes as speed rises. At lower speeds, well below the car's cornering limits, you could tell that you've early apexed by the need to turn the steer-

Fig. 3-1. Turning in early at the car's cornering limit and not realizing the mistake until reaching the apex curbing puts the car in a position—even if you do everything right from that point on—where it is going to end up off the racing surface. Recognizing the mistake earlier—in this case just five car lengths earlier—allows you to stay on the track.



ing wheel more in the second part of the turn. But as you become more aggressive about adding power through the apex and corner exit you'll find that you need to identify and cure the early apex much earlier.

Early Apex Diagnosis

Let's look at Fig. 3-1 to define the problem. If you did this corner perfectly you could drive through it in the range of 55 m.p.h.

If you turn three car lengths early, your arc into the corner is very broad, allowing you to corner at 67 m.p.h. on the way to the apex. If you reach point A before realizing that you've early apexed, you're in deep trouble: you're now going to ask the car to take a 33.5 m.p.h. corner at 67 m.p.h. It should be no surprise that the car continues on the big-radius arc right off the road. The initial problem was the early turn-in, but what guaranteed a bad outcome was realizing the mistake too late in the turn.

If you make the same initial mistake, but realize it earlier, the results are less severe. If you caught the mistake 62 ft. before the apex, straightened the steering wheel, and used the brakes hard, you'd be able to slow the car to 33.5 m.p.h. by point A, allowing the car to stay on the road at the corner exit.

This is a critical fact to keep in mind about driving race cars: The faster you go, the earlier you have to identify and cure mistakes. In this example you would have to identify and begin to cure the early apex over 90 feet before the proper apex of the corner.

When Mario Andretti was winning the World Championship in 1978, he had to give hundreds of interviews. In one, the interviewer, gushing superlatives, asked how it was to achieve perfection behind the wheel of a racecar. Andretti's reply was that no-one is perfect in a racecar. "The difference between the world class driver and the average racer," he said, "is that the champions recognize their mistakes sooner."

This type of skill calls for two things: one, knowing where the correct line is; and two, knowing where your car is on the racetrack, within inches, all the time. You obviously need reference points for your turn-in, apex and track-out. Also make yourself identify other reference points on the way into the apex, especially in long, fast corners where it may be a con-

siderable distance between the turn-in and apex. Notice where the car is in relation to these spots when you're going slowly on the right line. Then, as you go faster, any excursion *inside* these reference points on the way into the corner will clue you in to an impending early apex.

I Know It's Coming—Now What?

If the signs point to an early apex, you have to instantly switch your concerns from driving through the corner faster to driving away from the corner with the car in one piece.

You'll need to do two things. First, you need to admit to yourself that by going in early you have obligated the car to do more turning in the second half of the corner than it did in the first half—in short, you'll have to run a smaller radius in the second half. In order to do this, you'll need to be going slower. Job number one is to slow the car down.

If you're braking and turning into the corner, you could just rest your foot on the brake pedal a second or so longer than usual. If it's not a corner where you do a lot of braking and turning you could be more conservative on the application of power, providing balance for the car with some throttle application, but not accelerating up to the slowest part of the corner. If the situation were desperate, you could relax the steering effort, thereby asking for less cornering force, and use some braking to lose the speed required.

Second, you have to try to get the car back on the proper racing line as soon as possible—no later than the apex. If you succeed in having the car back to the right apex and at the right attitude at the apex, the second part of the corner will not be a tighter radius, and you can start picking up normal speed again.

What do we mean by the "attitude" at the apex? When the car hits the apex properly, not only is it close to the apex point but it's also pointing in the right direction. It is possible to make it close to the apex, yet not have changed the car's direction enough by this point in the corner. In Fig. 3-2, Car 1 is close to the apex but hasn't done enough change of direction for this point in the corner. The result will be the same as the early apex—more turning needs to be done later in the corner, requiring a tighter radius. Car 2 has got the apex and the attitude right.

Getting a reference point for the attitude of the car is a little more complicated than simply driving from dot to dot, but driving the proper arc over and over develops a mental image of what the apex and the corner exit should look like.

Using A "Sight Picture"

Many of our instructors refer to it as a "sight picture." After many repetitions through a corner, your mind develops a picture—a transparency, if you will—of the general scene in the corner. If the scene is different than normal, the car's position and attitude are

"If you're any good, the instant you turn your hands into the corner you know if you've made a mistake. You don't wait for the apex and say 'Oh gosh, I'm early or, I missed it by two feet'—way too late! When you see it early you have all that time to get out of the throttle a little and get the car headed in the right direction. If you're any good you make mistakes constantly but you never go off the road."

—Skip Barber

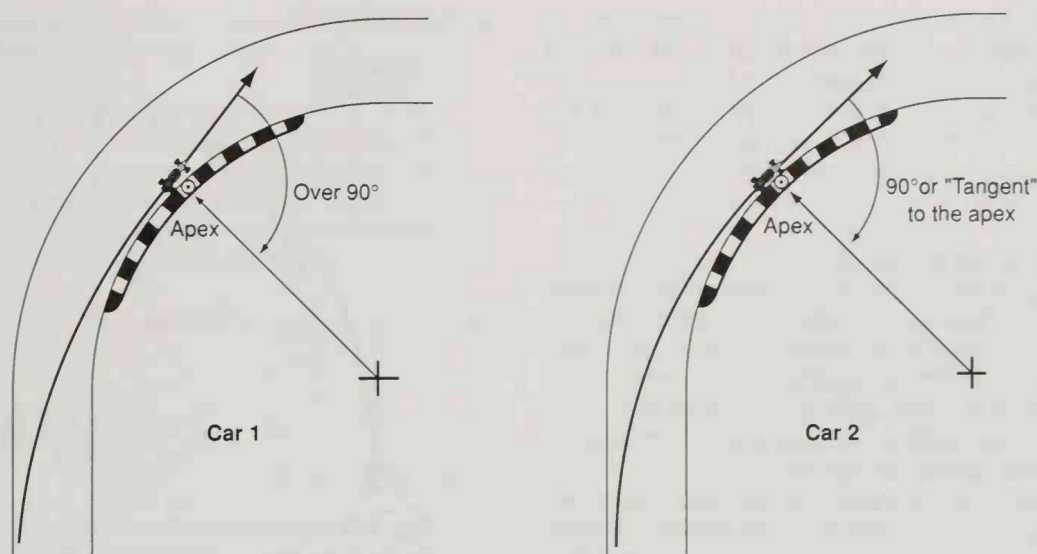


Fig. 3-2. Both these cars are near the apex but the car in the second illustration is at the correct attitude for its location in the corner.

wrong. The trick is to define how it's wrong and to decide on your next course of action.

Think of it as the same mechanism that is at work when six-year-old kids first try to catch a baseball. They know nothing of ballistics and trajectories, and their first attempts to predict where the ball is going to end up are futile. But as they do it more and more, their eyes and brains work together to make some predictions about what is likely to happen the next time the ball comes at them. After years of practice and thousands of repetition, good fielders start running to where the ball is going to be the instant the ball leaves the bat.

Through repetition, coordination, and athleticism, you can develop a sense of where you are in space versus where you need to be and how to coordinate your actions to get into the position you visualize.

Reference Points vs. "Feel"

While we've recommended orienting yourself on the racetrack using concrete reference points, it's worth knowing that some top-level professional racers don't. After 15 years of driving racecars, a racer is more equipped, with this level of experience under the belt, to use a sight picture approach to location on the racetrack. With less experience and fewer repetitions, the beginning racer is more likely to drive point-to-point. Starting with this approach and developing sight pictures over time is a common progression, but one is not necessarily any better than the other—it's all what works for you.

Early Apexing at Speed

When you get cooking, running through the corners very close to the limits of the car, the symptoms of early and late apexing change. We've looked at the kind of trouble an early apex can cause using some gross examples. Most drivers can get close to the right path, yet they always seem to make small line mistakes. An experienced driver is less likely to turn into a corner 30 feet early than the novice is, but in the process of refining the line at the limit, it's still possible to adjust the turn-in point too much.

Just like at lower speed, any need to turn the steering wheel more when past the apex is an obvious no-no. Mark Donohue always put a 1" strip of bright tape at the top of the steering wheel so that he could clearly see which way the steering wheel was heading on the way out of the corner. If it was heading toward the inside of the turn, the turn-in and apex were too early.

At racing speed, you should always be unwinding the steering wheel at the corner exit. If not, try a later turn-in and apex. There are, of course, exceptions. If the car is at terminal velocity, that is, there is no acceleration going on, then it's okay to have a stationary

"I start with fixed reference points, which I think you need to identify the line, but, as the weekend goes on and you've been through the corner more and more, the visual picture becomes implanted in my mind and the fixed points start to go away. You end up at those same points but you are not concentrating on them, per se, any more."

—Brian Till

steering wheel at corner exit. In the vast majority of cases, if you can't unwind the wheel, you're too early.

Throttle position is another good indicator. If you have to breathe off the throttle coming out of the corner you may not have left yourself room enough to widen the arc at the exit. Solve it by turning later and apexing later.

Late Apexing at Speed

At racing speed the late apex provides you with a dilemma. The later you turn, the more change of direction you accomplish in the first part of the turn—and the less you have to change the car's direction at the exit. This makes putting the power on easier and makes it easier to unwind the steering wheel: both are goals you'd like to accomplish.

The catch is that when you turn later, you have to go slower at corner entry. In our earlier Turn 7 example, turning one and a half car lengths too late (20 feet) cut your potential speed to the apex by 5 m.p.h.

Some cars will make up that 5 m.p.h. loss and then some by the track-out point. Some racecars take *forever* to gain back 5 m.p.h. You could be late apexing the corner, but since you're unwinding the wheel and the car is ending up at the edge of the road at the track-out point, you think you're doing fine. In fact, you could be losing time because, for your car, the late apex costs both cornering and exit speed.

How do you tell? It should be a bit of a struggle, even though the steering wheel is being unwound, to exit the corner without dropping a wheel off. If it's too easy perhaps you're giving away too much entry speed by taking too late an apex. A valuable technique to get into the habit of using is to check the tachometer (or speedometer, if your car has one) at the exit of every corner. In trying to decide the turn-in and apex to use, start with a late apex, then begin turning slightly earlier and look for an RPM improvement at the track-out. If the exit speed is improving, keep moving the turn-in earlier until symptoms of the early apex show up.

SHORTCUTS TO THE LINE

It would be wonderful if we all could have unlimited track time to practice, practice, practice. We don't. Luckily, there are things you can do *off* the racetrack to make yourself a better driver. When it comes to finding the line, you can make some generalizations that can help you find the right path around the course in less time.

Constant Radius Corners

A corner which can be defined by one radius, like Turn 7 at Sebring, is a constant radius turn (see Fig. 3-3). Here, the inside edge of the road has a 100-foot radius which doesn't change from the beginning of the corner to the end. In such a corner the apex is close to the midpoint, around halfway from turn-in to track-out.

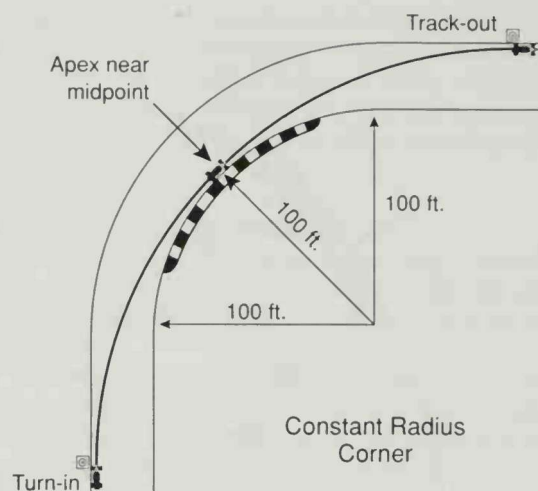


Fig. 3-3. In constant radius corners, start looking for the apex around the midpoint of the turn.

When you first go out on the circuit, don't waste your time starting with a late apex 7/8 of the way around the corner. Look for the apex slightly later than the midpoint and use your knowledge of symptoms to refine your reference points.

Constant radius turns aren't restricted to 90 degrees. They can be from 10 degrees to about 120 degrees (over about 120 degrees, a corner becomes a hairpin turn).

Decreasing Radius Corners

A decreasing radius corner is one where the radius of the later part of the turn is smaller than that of the entry (Fig. 3-4). In these types of corners the apex is further along than the halfway point—a later apex. If you

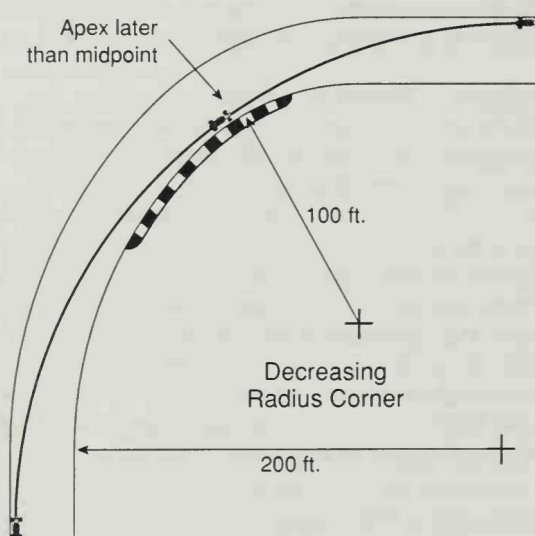


Fig. 3-4. If the corner has a decreasing radius, look for the apex later than the halfway point.

thought it was easy to early-apex constant radius corners, wait till you try one of these.

Increasing Radius Corners

If the radius of the first half of the corner is smaller than that of the exit, you've got an increasing radius corner on your hands, (see Fig. 3-5). The apex will be earlier than halfway through the corner and the turn-in will be earlier than you might guess.

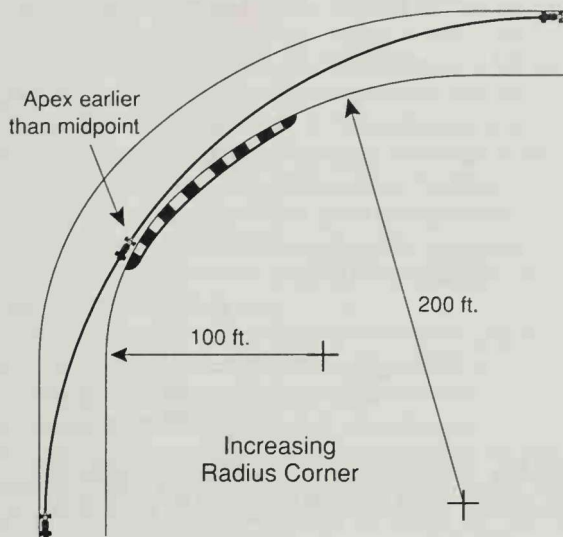


Fig. 3-5. In increasing radius corners, the apex is earlier than halfway through the turn.

Hairpins

When faced with a corner like this, you've got plenty of choices. You could apex it at the midpoint. If you did just that, 90 degrees into a 180-degree corner (Fig. 3-6) with a 100-foot radius, you could get a 130-foot arc good for 44 m.p.h.

The primary problem here is not so much the speed but that with a single constant radius, you can't increase speed at the corner exit. Consequently, exit speed is the same as entry speed: 44 m.p.h.

You could create a bigger radius if you treated the corner less like one 180-degree change of direction and instead divided it into two 90-degree corners. You could create a fast (over 55 m.p.h.) arc into the corner and another fast one out (see Fig. 3-7).

The problem here is that you need to "park" the car in the middle part of the corner to get its direction changed. "Park," in this case means slowing to 30 m.p.h. or so. If the big arc can support 55 m.p.h., you could come out of the corner 11 m.p.h. faster than your first approach of apexing the corner in the middle. To take advantage of this approach, you need a car that's capable of going from 30 m.p.h. to 55 m.p.h. quickly. The longer the car takes to accelerate from 30 to 55, the less likely it is to gain an advantage from apexing the hairpin twice.

A good Indy Car car, for example, can accelerate at over 1G—that's 22 m.p.h. per second. It will go from 30 to 55 just fine, thank you. A Formula Vee, on the other hand, powered by a 65-hp motor wouldn't be able to gain back speed at nearly the same rate that it lost it. The wiser thing for the driver of a low-horse-

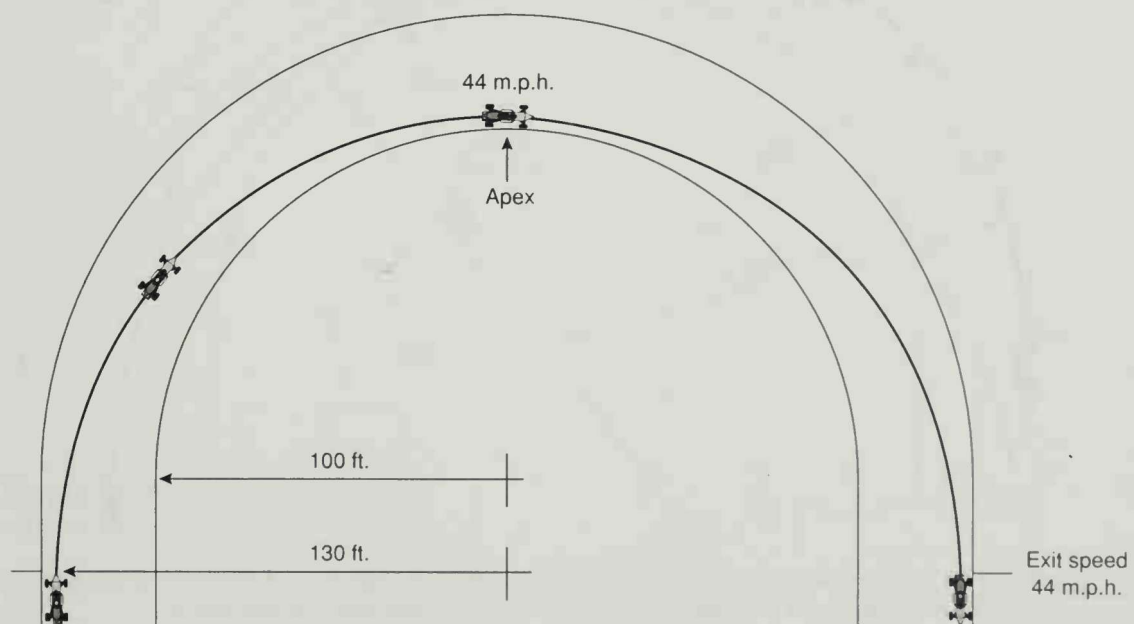


Fig. 3-6. Hairpin corners create dilemmas. Taking a single apex halfway through the turn limits the car to a long period of constant speed cornering.

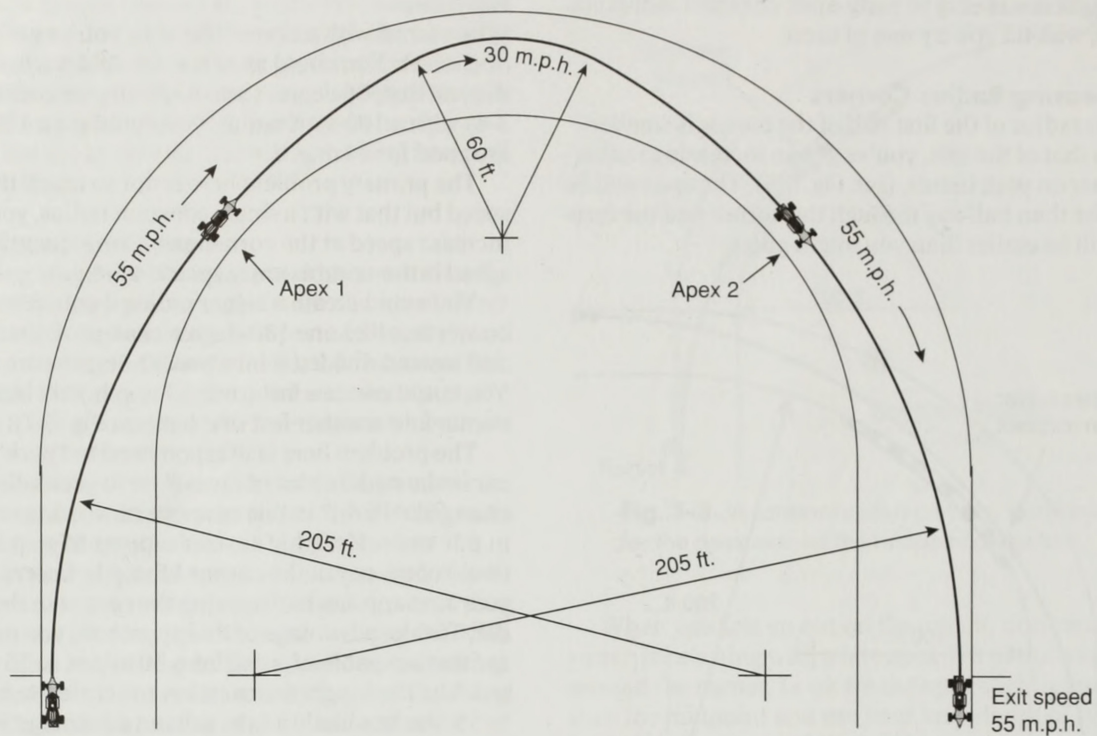


Fig. 3-7. An alternate approach to hairpins is to form two apexes, resulting in potentially higher entry speeds and exit speeds, but often creating a mid-corner section where speed is limited by a tight radius.

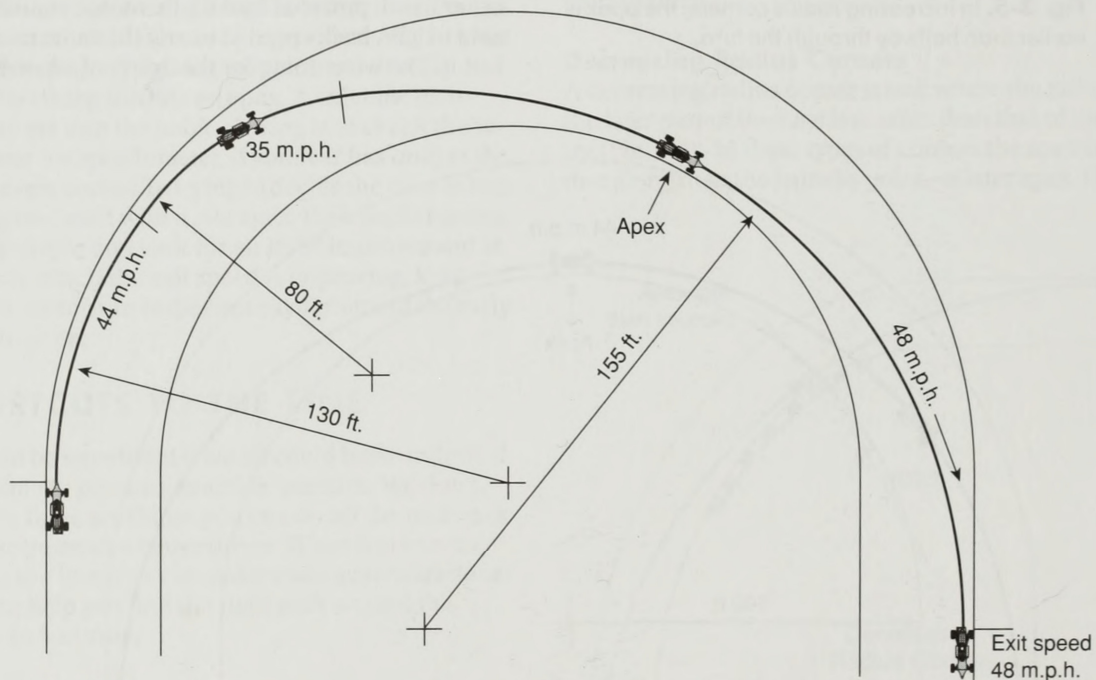


Fig. 3-8. The most common approach to hairpins, especially tight ones, is to form a single late apex that allows higher corner exit speeds, critical if the corner leads onto a long straight.

power car would be to avoid any extreme tightening of the arc. Instead of double apexing the corner, it would be better to begin the hairpin on an arc that would keep the speed up but still allow for an increasing radius arc out of the corner.

The most common approach to hairpins, especially tight hairpins, is to take a single late apex about 3/4 around the corner (see Fig. 3-8). Some speed is sacrificed at the entry of the corner but not as much as is lost by using the double apex. The concentration is put on maximizing exit speed and starting the run out of the corner as early as possible at a higher initial velocity.

That doesn't mean you don't double apex 180-degree corners unless you've got more than 600 hp on board. If the 180 is broader, with a radius of several hundred feet and up, the change in direction in the high side of the corner is more subtle, requiring less of a speed loss. Cars other than great accelerators can then take advantage of the entry speed advantages of the double apex without putting themselves at a great disadvantage coming out.

In the 450-foot radius corner pictured in Fig. 3-9, the speed difference required to change the car's direction back toward the second apex is down to 18 m.p.h. compared to a 25 m.p.h. change required in the 100-foot radius example we used earlier. As hairpins get bigger and bigger radii, they begin to fall into the category of "carousels," and the speed change required to get turned down to the second apex gets smaller and smaller.

"The size of the radius of a hairpin is probably the biggest part of the decision to single or double apex. I can make up my mind pretty quickly by reading how sharp or broad the corner is. If it's fairly tight, I'm going to assume it's a single apex. If it's broad and fast, you try two and see how it works."

"The way I handle a hairpin sometimes depends on the car I'm driving. If the car doesn't have a problem with the direction change in the middle of the corner then I'll try to blast the entry a little bit and slow down in the middle—treat it like an oval where you come in hard, gather it up in the middle, and then come out hard. The Group C car, for example, was not very nimble and didn't respond well to a mid-corner line change so, with that car, you avoided double apexing a hairpin."

—Jeremy Dale

Notice in this illustration that the line in the high part of the corner doesn't go all the way to the outside of the road. The further into the corner you take the original broad arc, the more drastic the change of direction necessary to get the car headed toward the second apex and consequently this portion of the corner has to be slower.

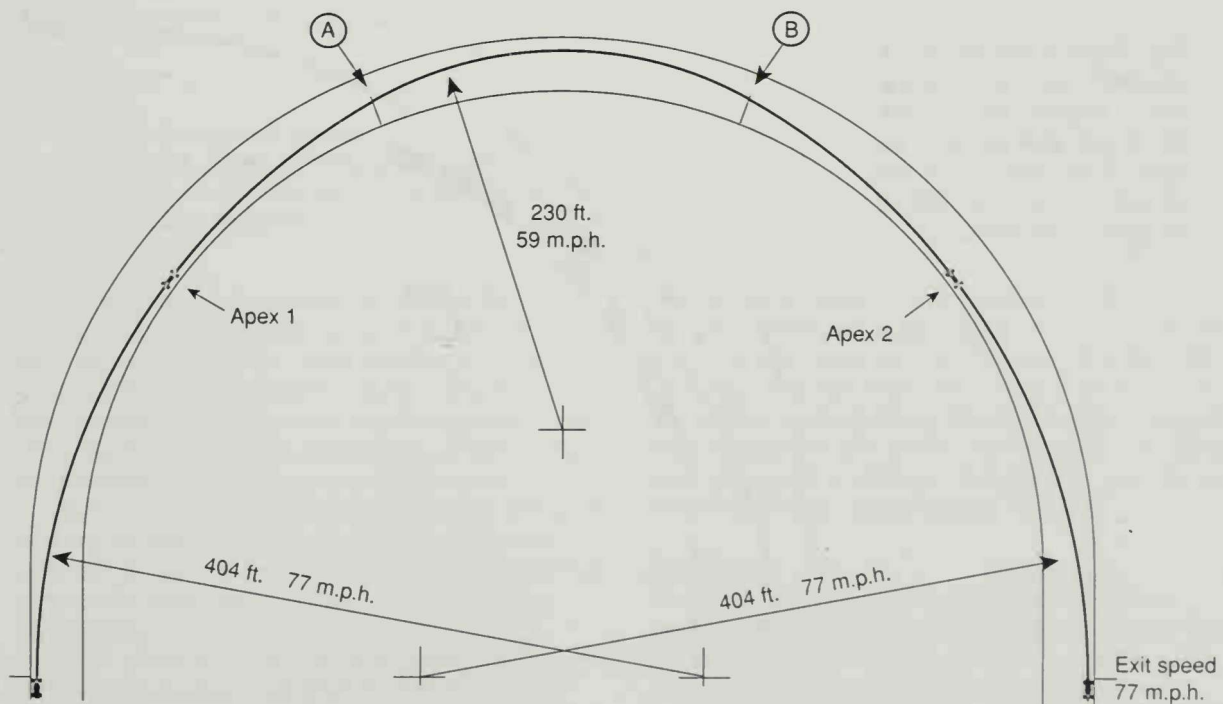


Fig. 3-9. As the radius of the hairpin gets larger, a more subtle radius change in the middle of the corner between point A and point B makes the double apex approach more sensible.

In these diagrams we've been giving radius and speed numbers just for illustration purposes. Clearly, the car doesn't lose speed instantaneously as it moves from one radius to another. What's really going on is that, depending on how you're using the controls, the speed and radius change simultaneously and gradually. On the way into the corner, the speed is going down and the radius is decreasing. At the corner exit, the opposite is happening—you're accelerating and the radius is expanding.

BEYOND GEOMETRY

The shape of corners is going to give you hints about approximately where the apexes are going to be but, especially at speed, the line complications don't end here. When looking for the optimum line, you need to determine where in the corner the car is going to change direction best.

Changing Traction

Generally, in a constant radius corner you'd like to do half of the car's direction change before the middle of the corner and the other half of the direction change after the apex.

This works out fine if the car changes direction as well in the first part of the corner as it does in the second part (see Fig. 3-10). In our chalkboard corner, where you can only see the corner as a space between

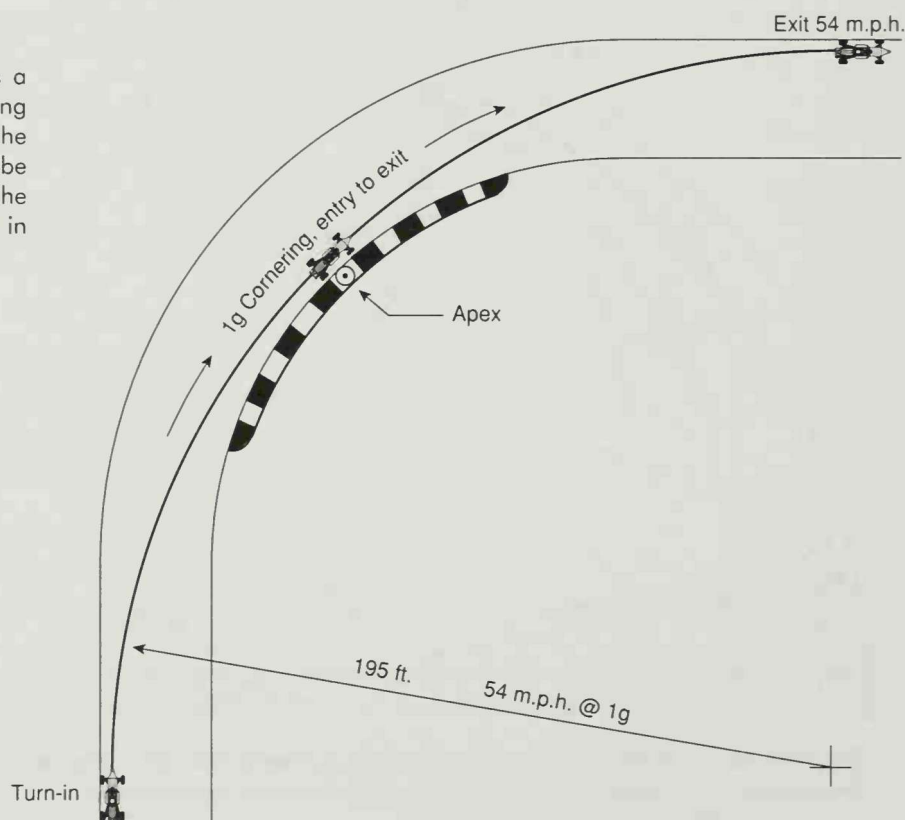
two lines, this seems like a reasonable assumption to make. In real life, however, it's not that simple.

If, for example, the car generates twice as much cornering force in the second part of the corner than it did in the first (Fig. 3-11) it allows you to turn in earlier and carry more speed into the corner on an early-apex line: you can use the superior turning ability late in the corner to change the car's direction and keep it on the road, all at a higher speed than you might think possible.

If, instead, you take the constant radius arc that *appears* to be right based solely on the shape of the corner, you have to go slower at the entry of the corner because of the smaller radius arc that's required to get to the apex—in this case, 13 m.p.h. slower! Turning too late in this instance obligates you to a 54 m.p.h. arc as opposed to a 67 m.p.h. arc if you read the corner correctly.

Here's the general rule for variable cornering force turns: If the car corners better in the *second* half of the corner than the first, the turn-in and apex are likely to be *earlier* than the shape of the corner would dictate. If the car corners better in the *first* half of the corner, (see Fig. 3-12) than the second half, the turn-in and apex are likely to be *later* than the shape of the corner would dictate. In this situation you would use the additional grip to get more of the car's direction changed early and to reduce the amount of turning required in the later part of the corner.

Fig. 3-10. If the car has a consistent level of cornering force, roughly half of the change of direction can be done in the first half of the corner and the other half in the second half of the turn.



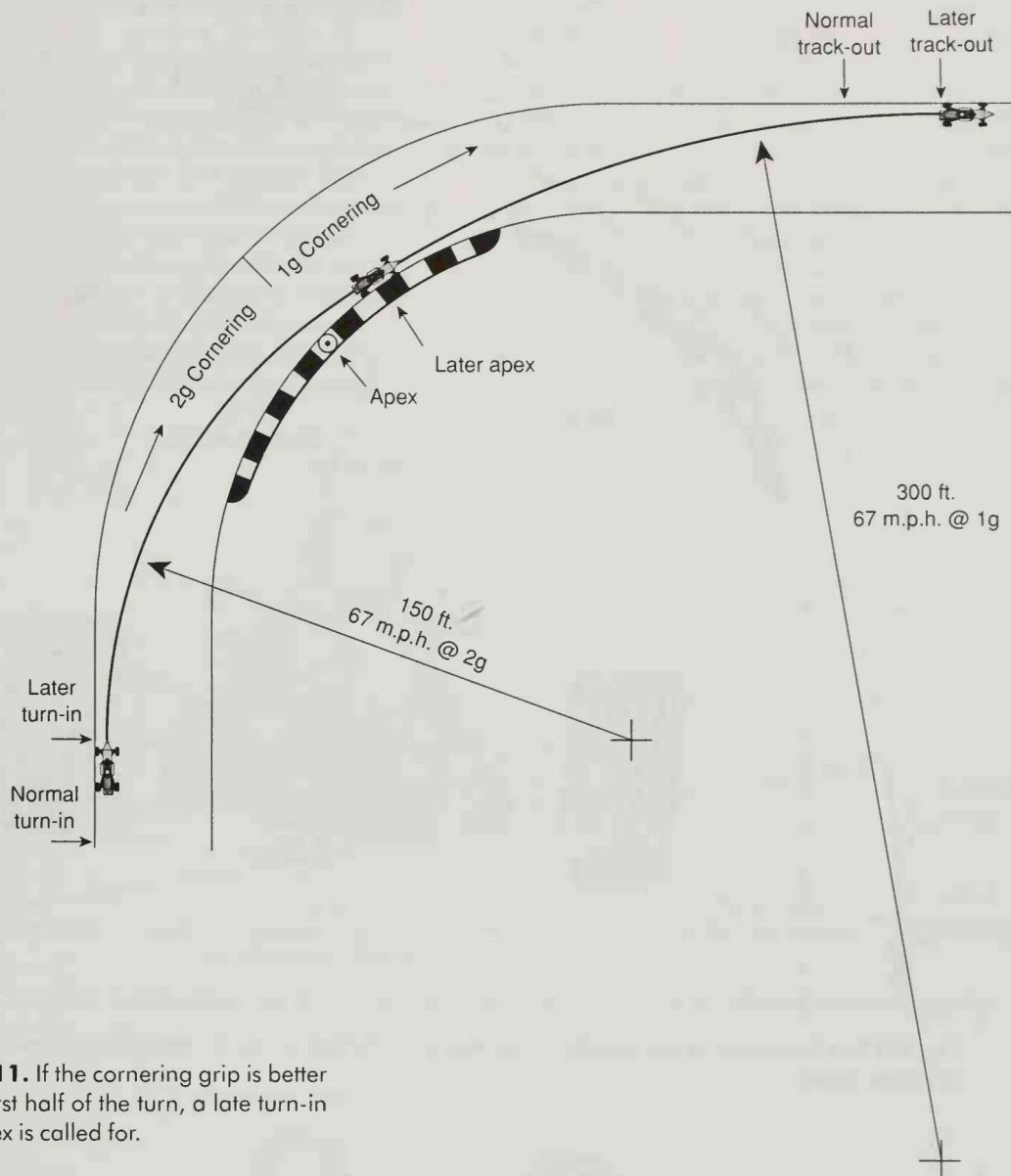


Fig. 3-11. If the cornering grip is better in the first half of the turn, a late turn-in and apex is called for.

There are many things that can change the car's cornering force *during* the corner. Anything that affects the tires' grip on the road qualifies as an influence and has to be considered in choosing the best line through the turn. A short list of factors includes the composition of the racing surface, debris, track temperature, bumps, elevation, and banking.

Racetrack surfaces are seldom perfectly horizontal and any variation from level is going to affect how well the car corners. Let's look at the cornering forces at work on a flat road and then look at how they change when the road is banked.

Fig. 3-13 shows a 1,000-lb racecar turning to the right at its cornering limit. If this were a perfect 1 G racecar, all four tires would be exerting their maximum sideways force, a total of 1,000 lbs, to the right (represented by the bold arrow). Resisting this turning force

is the force of inertia, which essentially wants everything to remain at constant speed or direction. When you force your car to alter its direction, inertia, in the form of centrifugal force, resists that change with an equal force in an opposing direction (represented by the fine arrow pointing left). On a flat road, 1,000 lbs of force generated by the tires is resisted by 1,000 lbs of centrifugal force—everything is in balance.

Banking

But when you change the surface of the road to a banked surface, the balance of forces also changes. Let's put the same 1,000-lb car in a corner where the road is canted 5 degrees toward the apex, (see Fig. 3-14).

The first thing you need to determine is how hard the tires are now pushing down on the road. As you'll learn in detail in the chapter about tires, there is a di-

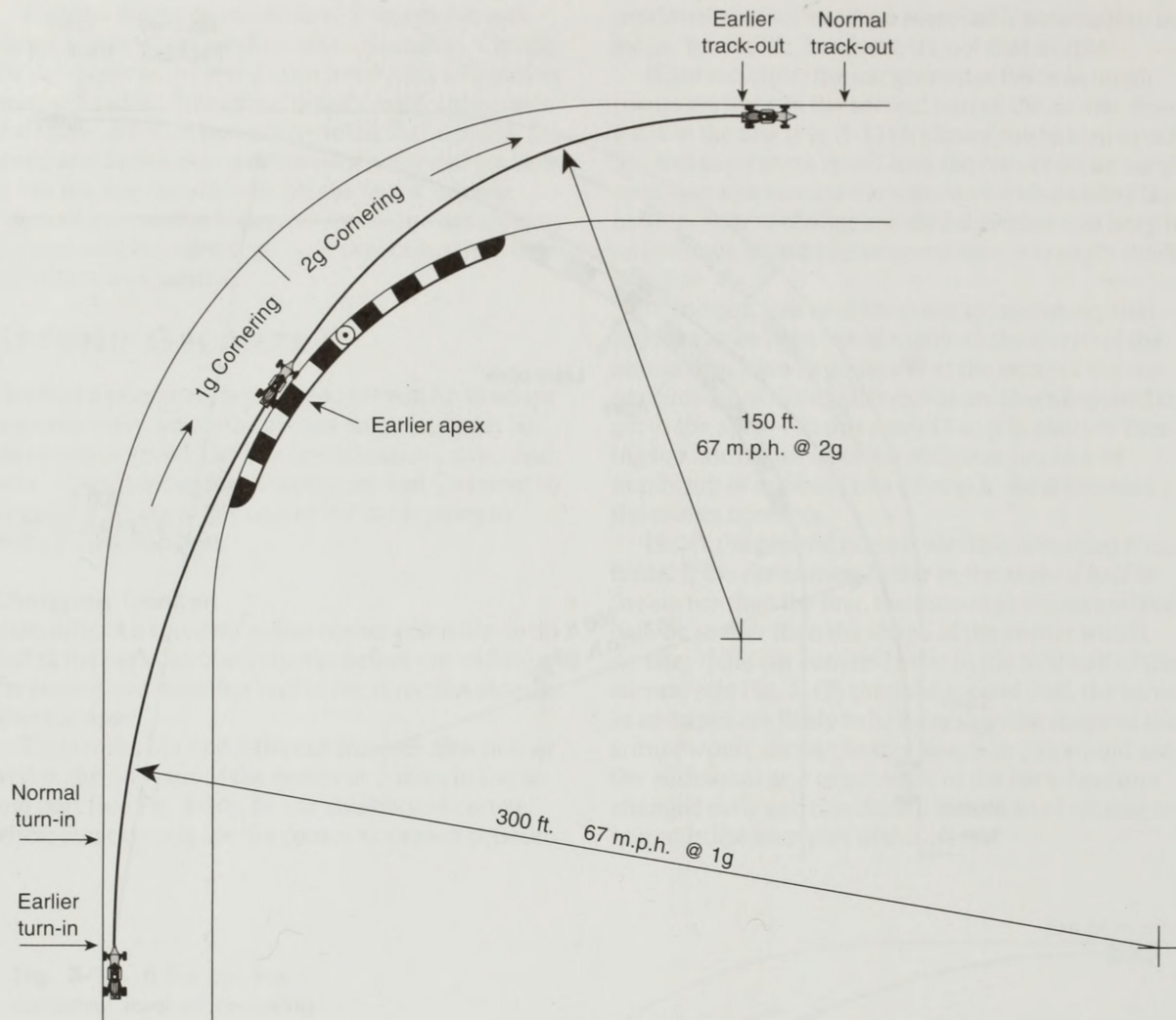


Fig. 3-12. If the cornering grip is better in the second half of the corner, an earlier turn-in and apex can result in higher speed.

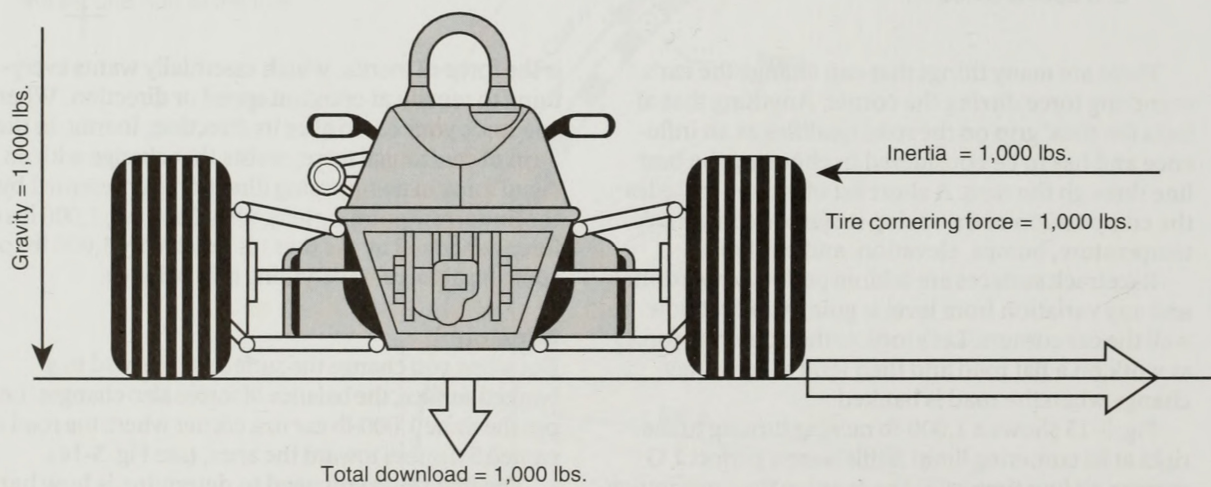


Fig. 3-13. On a perfectly horizontal track surface, the downforce on the tires is purely a function of gravity (excluding aerodynamic effects).

rect relationship between download on the tires and cornering force.

On a flat road, the four tires on this car had 1,000 lbs of download on them and delivered 1,000 lbs of cornering force—a realistic expectation. On 5-degree banking, the car's 1,000 lbs is no longer *all* going toward loading the tires. Using a little trigonometry, you'll find that at 5 degrees, only 996 lbs of the car's mass is loading the tires. Another part of the thousand-pound load, the part represented by the arrow parallel to the road's surface, will actually contribute to the forces acting to change the car's direction.

When you set the car in motion things change again. If the car goes through the corner at a speed

which creates 1 G of cornering force—54 m.p.h. in our 195-foot radius of Sebring's Turn 7—the centrifugal force no longer acts in direct opposition to the turning as it did on a perfectly horizontal road surface. Part of the centrifugal force (represented by the bold arrow in Fig. 3-15) is diverted into loading the tires. This force has two components: the downward component perpendicular to the road, which increases grip and limit speed; and the component resisting the change of direction parallel to the road.

On a 5-degree banking in a corner like Turn 7, centrifugal force eventually contributes over 100 pounds of downforce to the tires of a 1,000-lb racecar, providing over 10% better cornering force. That's a lot. Race

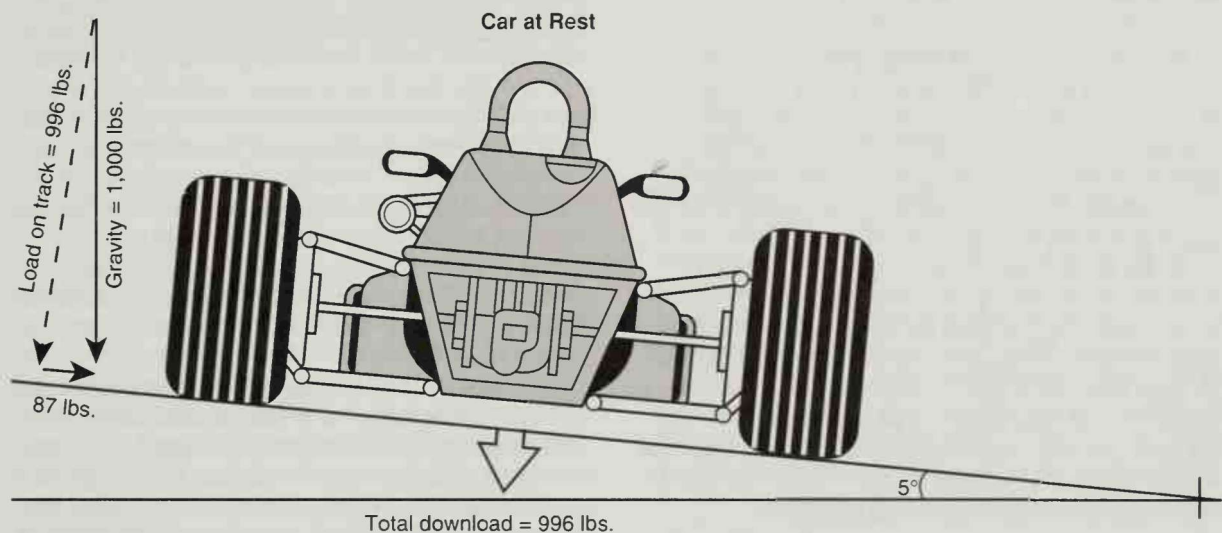


Fig. 3-14. When the road surface is banked, even subtly, the download changes, affecting cornering ability.

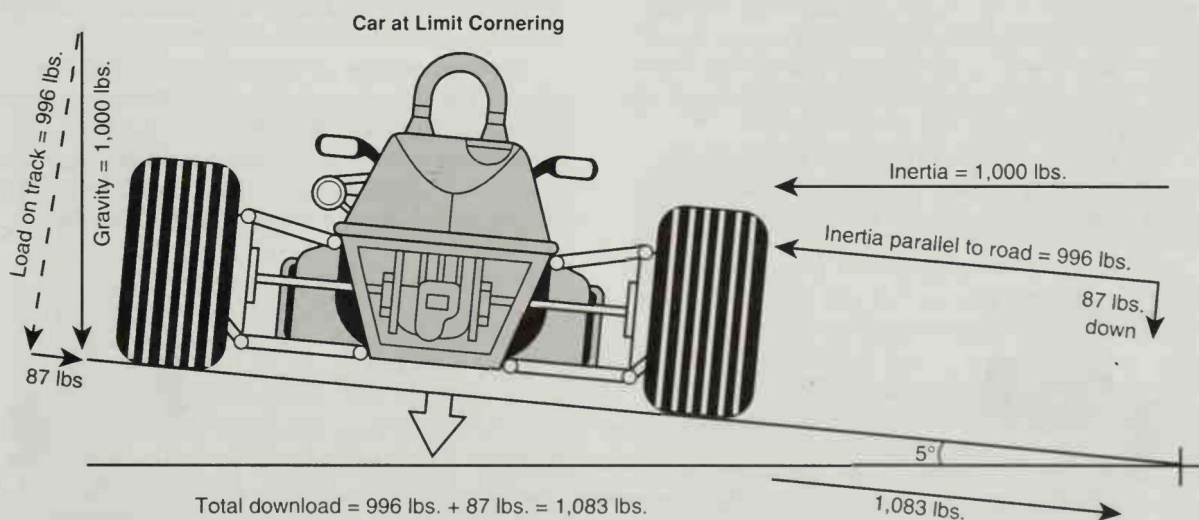


Fig. 3-15. On a banked track surface, a number of effects combine to increase the car's ability to change direction.

car designers would sharpen their mechanical pencils with their teeth for 10% more cornering force. Advantage Number One of banking is that it creates more traction by loading the tires more.

Advantage Number Two is that gravity is helping the car change direction. With a 1,000-lb car, 87 lbs of force is acting parallel to the road surface to help pull the car toward the apex. Picture the road covered with ice. That 87-lb force would move the car sideways from the top of the banking right down to the bottom.

Banking, even subtle degrees of banking (or "road camber"), has a significant effect on the car's cornering limits and the speed at which you can negotiate portions of the circuit. For instance, in our Sebring example, if the corner were banked 5 degrees in the later portion of the turn, your 1 G car could go 59 m.p.h. there as opposed to 54 m.p.h. on a flat corner.

By changing your line so that you could arrive late in the corner at the higher speed, i.e. by turning in early, you could improve both the corner time and the exit speed onto the straight. There are potentially great advantages for noting even subtle changes in road camber. Even one degree of road camber can improve cornering grip by over 3%.

So far we've focused on the positive aspects of road camber, but it's quite common for road courses to have banking that goes the wrong way: what's commonly called "off-camber." If this happens, the double gain that makes banking so wonderful becomes a double loss as the car loses significant traction. The car won't become uncontrollable: it just acts like a car with a lot less grip and will slide around more than expected and consequently use up more road.

Multiple Camber Changes

Sometimes you'll end up with multiple changes in road camber in the same corner. While ovals and some purpose-built road courses are very carefully laid out and surveyed by engineering crews, many older courses were simply paved over the lay of the land. There may or may not be consistency to the road camber from entry to exit of the corner.

The classic example of this is on street circuits where the roads are laid out to accomplish different goals. Most public highways are crowned, higher in

the center than on the edges, to help drain rain water toward the gutters. This creates problems for racers that have to negotiate a corner where a cross-section of the road surface looks like it does in Fig. 3-16.

At the turn-in point, the road is off-camber, reducing cornering force at the entry. As the car crosses the middle of the road, the situation improves until at the apex everything is wonderful. Coming away from the apex, the car loses grip as the road camber goes away, ending up at the track-out with a lot less grip than it had just seconds ago at the apex. It's your job to be aware of the different factors that will affect the race-car's performance. Noting road camber changes is part of that job.

Many drivers don't make the effort and the following scene is played out far too often at circuits that have crowned road surfaces. Driver and team set out for a practice day at the unfamiliar circuit to "set the car up." The driver goes out and in ten laps comes in and gives the crew chief the following report: "I dunno. On the corner leading onto the pit straight I can't seem to get the car turned in—it just won't point. Once I manage to get it across the road it's just great at the apex but as I come off the corner on power it skates to the outside and it wants to fly into the weeds at the exit." The crew chief who makes chassis adjustments to solve this "problem" engineers his way to the back of the grid.

On public highways, the typical crown in the road is 4 to 6 inches higher at the center than at the curbs. This would represent at least 1-1/2 degrees off-camber at the turn-in and track-out and 1-1/2 degrees of positive camber at the apex. With a 1 G racecar, the apparent cornering force would vary from .956 G to 1.044 G over the course of the corner. This initially doesn't sound like a big difference but it means that you have 4.5% more traction at the apex and 4.5% less at the turn-in and track-out. That's a 9% spread—something that needs to be considered.

Even permanent circuits need to be considered closely. On "The Downhill" at Lime Rock, a fast right-hander leading onto the straight, the road has 1 degree of road camber at the turn-in. What complicates things is that the turn-in is at the base of the hill. The car becomes heavily loaded, creating gobs of grip—

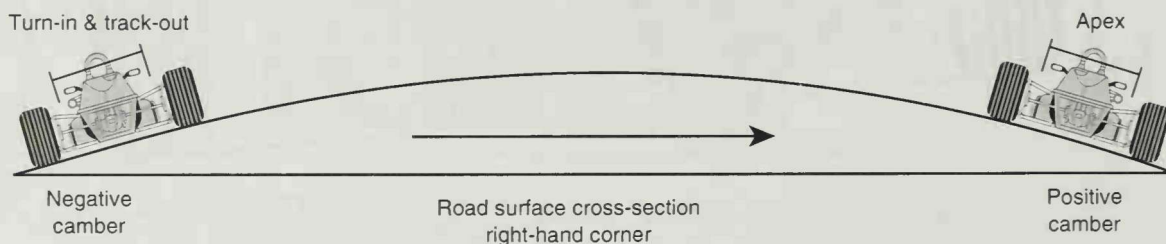


Fig. 3-16. Multiple road camber changes throughout the course of a corner can substantially change the car's cornering ability and the driver's choice of line.

"The line can be a difficult thing to teach because it gets complicated, and there's a lot of feel to it. The standard geometric line might not be best because the surface may be bad or the camber is wrong, or there's a crown in the road.

"You've got to go where the grip is. In the Keyhole at Mid Ohio, for example, you'd think you should start out wide, then come in on the apex. We all tried it like that to start, but there was no grip out wide—it was off-camber—so what worked there was just to stay down on the inside to the apex. Doesn't look right, but you've got to be able to learn the line, then adapt to what the track gives you."

—Danny Sullivan

for the moment. The road camber increases as you approach the apex, reaching a maximum of +4-3/4 degrees. Coming onto the straight, the banking falls away to +1-1/4 degrees at the middle of the road to -1-1/2 degrees off-camber at the track-out.

Road Surface Changes

It is easy to assume that the entry of the corner is made of the same stuff as the exit, but you really have to look. Asphalt is expensive, so when it's time to repave, often the racetrack will do the worst sections of the circuit first, sometimes resulting in a corner that has one type of asphalt at the entry (aged, weathered and leaked on) and brand new stuff from the apex out. If the pavement surface will affect the car's traction, it will affect the line. Drivers run into this consideration more often these days with the proliferation of street circuit races. I can recall counting eleven different pavement surfaces at the Meadowlands one year—four different types in one corner alone.

"There's different amounts of grip on different types of pavement surfaces and a lot of it depends on the conditions. The best example, I think, is Mid Ohio where the track is asphalt but there are several cement patches in the middle and exits of corners. Depending on the weather conditions, there's different amounts of grip on those different surfaces. When it's very cold, the asphalt generally has more grip than the cement, so I might take a later, wider, entry in order to get to the asphalt quicker or to spend more time on the asphalt. When it's hot, the asphalt loses grip, where the cement, because it's a lighter color, stays cooler and so it has more bite."

—Bryan Herta

Bumps

For the tires to deliver traction they need to be in contact with the road surface. If the surface of the road at the geometric apex of a corner is terrifically bumpy, causing the tires to be in the air half the time, perhaps staying out of the bumps would create more potential cornering speed than the extra few feet of radius would. Similarly, if the exit of the corner is bumpy, making it difficult to accelerate and turn off the corner, getting more of the direction change done earlier in the corner may help.

Elevation Changes

Changes in elevation during the course of the corner have much the same effect on the line as banking. As a racecar runs into a hill, inertia creates a download on the tires, increasing the tires' traction phenomenally. At our home track at Lime Rock we have one of the classic elevation change corners in the United States in "The Uphill."

To give you an appreciation of the downforce that the Uphill can create, we could have four of our mechanics stand on the rear bulkhead of one of our Formula cars, and their combined weight of 700 lbs would not make the car bottom. But, because of the compressive force of the road going steeply uphill, we have been forced to put skidplates on all of our cars to keep from grinding the bottom of the chassis away.

This downforce creates substantial grip late in the corner and changes the line. As you can imagine, the opposite effect is just as critical. If the car comes over a crest, as it does before the exit of the Uphill, it will reduce traction to nil in some cases. You'd better not be trying to turn the car, and you'll sometimes find that getting the steering wheel dead straight over the crest of a rise may be required.

Slippery Conditions

In most of the situations we have been dealing with you can anticipate the change in the line through the corner by doing some advance homework—looking for road camber changes, differing pavement surfaces, varying shapes of corners. Often, however, the need to make an alteration to your standard line comes up rather suddenly. Anything that can make the racetrack slippery can affect the line. The list includes sand, mud, dirt, leaves, water, anti-freeze and—the worst—oil. Rain is a special sort of slippery condition which we'll cover in detail in the chapter about rain driving.

Say, for example, you are coming up to Turn 7 at Sebring and, looking to the left as you approach your braking point, you catch out of the corner of your eye an unusual motion followed by a cloud of dust. You have four seconds to gather the facts, plan and execute your response. Clearly, you have to be prepared to alter the line through the corner, but before you even turn the steering wheel, your response should be



Fig. 3-17. Elevation changes can increase or decrease grip dramatically. Here, Jeremy Dale lifts the front tires off the ground at the top of the "Uphill" at Lime Rock.

to slow down. It sounds almost too obvious to mention, but it's a fundamental truth about racing. Racers are intent on maximizing their speed at every opportunity and tend to forget that sometimes it's wise to slow down when things start to go badly.

Many novice drivers overreact and slow the car to a crawl, often creating more havoc behind them as other drivers stack up accordion-style. A car that can brake at 1 G (a pretty common racecar) can lose 2.2 m.p.h. per tenth of a second, so braking just a tenth of a second earlier can put you at the turn-in point over 2 m.p.h. slower than a normal lap. Two tenths of a second earlier gives you almost 5 m.p.h. of cushion.

Sometimes you don't see the incident that caused the traction change. Here's where good corner workers can save you. A yellow and red striped flag displayed at the entry to the corner means that there is either debris or a slippery condition in the corner. Don't second-guess the flag. It's not a command to slow down, but it's valuable information and you'd be a fool to ignore it.

Once you've handled the braking, you have to decide how to modify your line. If the traction is affected more in one half of the corner than another, the alteration of the line would follow the same guidelines you used in variable cornering force turns. The only difference is how fast you need to decide.

Most of the time, the line change in these situations is to apex the corner later. Most spins have their conclusions at or past the apex, so you're more likely to see sand and dirt and lose traction late in the corner. Occasionally, drivers *will* spin under braking, mucking up the first part of the corner. The correct re-

sponse would be to stay inside the muck if you can, early apexing. Going in early is going to require more turning late in the corner, so slow down.

In conditions where the track "lubricant" has come from another car (anti-freeze, oil, or a combination), how it affects the line depends upon whether the car dumped all its fluids in one place or dribbled it out a bit at a time, coating the entire racetrack (on the line, of course).

If it's all in one corner, you make a line adjustment based on where the spill is. If it's all over the racetrack, you'll need to alter not only the line in the corner but also the braking zones on the approach to the turn. Either way, you have to switch from maximizing your radius to maximizing your grip.

You'll want to slow for the corner on dry pavement, and the greater the speed loss required for the corner, the more important in terms of lap time this will be. At some point you'll have to cross the greasy stuff, but you'd like the crossing to be as quick as possible, so you may drive diagonally toward the outside, making as perpendicular an intersection as possible. Expect that the first turn of the steering wheel into the corner may not change the car's direction much—the tires will pick up some coating as they cross the slick. As it wears off, the car will bite better and better.

The next task is to get the car's direction changed and accelerate away from the corner, knowing that you'll have to cross the slick again. This is one of those times when reading the tach at the corner exit becomes a big help as you try different paths to see which yields the greatest exit speed. Use the dry, abrasive pavement to get the turning done, straighten the

wheel as much as possible, and get across the greasy stuff as quickly as you can, modulating the throttle all the while to avoid excessive wheel spin. It ain't pretty.

Changing Back

Once some substance makes the racetrack slippery, it doesn't necessarily last forever. Dust and dirt blow away quickly, especially if there is a big field of cars racing, so your alteration to the dry line may last only a few laps.

Water and coolant tend to stay around longer, but on a hot, sunny, windy day, it too may have a short-lived effect. Oil is pretty persistent: it hangs around for a long time. However, we've seen racetracks that were undrivable get back within one to two percent of their pre-spill lap times within a half hour. As the conditions are changing, the good drivers are gradually working their way back to the dry line using the symptoms of early and late apexing and reading their exit speeds to define what the best line is under present conditions.

THE WHOLE RACETRACK

So far we've been looking at each corner as if it were the only turn on the course. You can't afford to make this assumption. In fact, not all corners were created equal. A particular corner's placement on the course, where it leads to and what precedes it, may also affect the line you choose to drive through the corner.

In Fig. 3-18, you have two simple constant radius 90-degree corners. Choosing the biggest radius arc through the first would make it impossible to take the biggest radius arc through the second. What to do?

Compromise Corners

In order to answer the question, you need to decide which of these corners, the right or the left, has the greatest effect on your total lap time when driven on the biggest radius arc.

The first thing you need to know is where the second corner leads. If it leads onto a long straight, you would like to carry as much speed as possible away from the corner and minimize the time it takes to get to the braking zone for the next turn. This is the case here. Given the choice to drive on the big arc in the

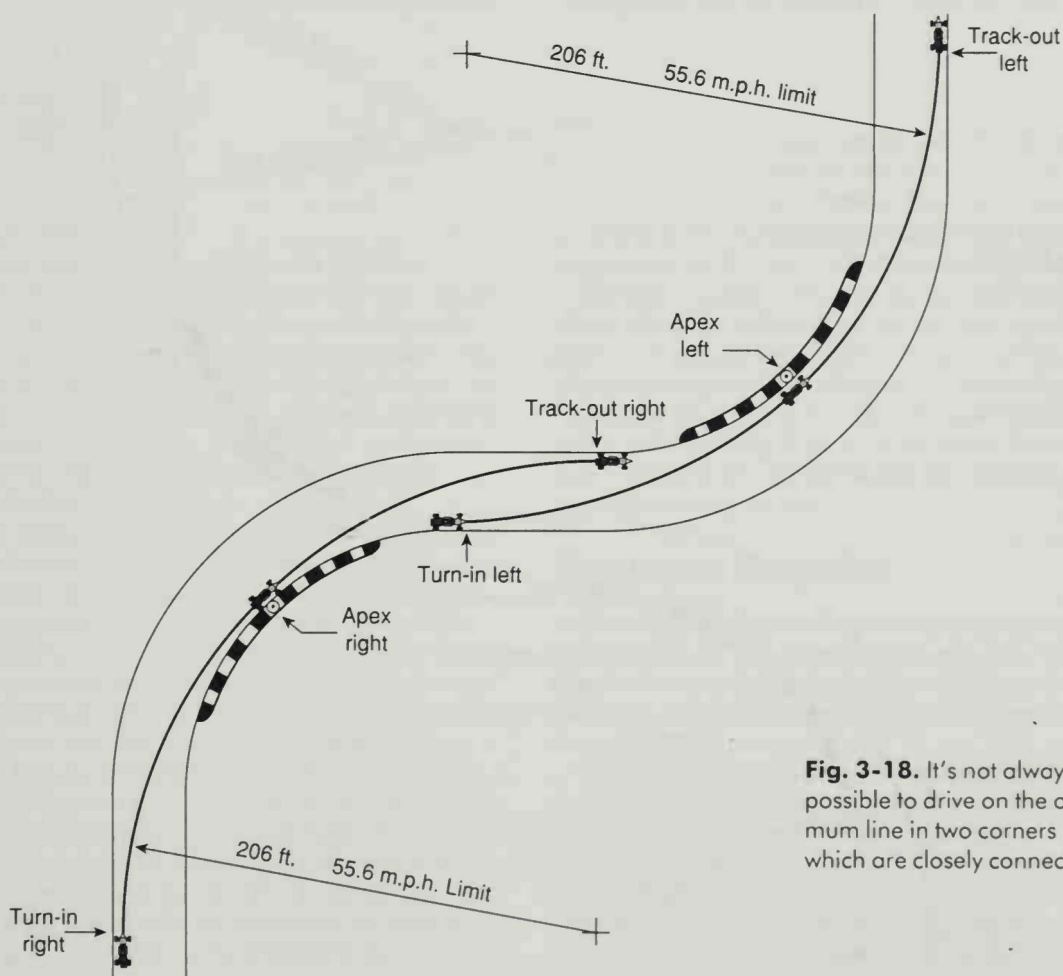


Fig. 3-18. It's not always possible to drive on the optimum line in two corners which are closely connected.

"Once you realize a corner is a compromise corner, then it turns into a discipline corner. I almost always find that it's easier to go faster on the wrong end. The fact that you could do the first corner really fast will make you feel like you're quick—but then you blow the second corner and subsequently you're down the straightaway slower."

—Dorsey Schroeder

right and a smaller arc in the left or a smaller arc in the right followed by a big arc onto the straight, the latter choice is the best.

In this case, you make a conscious decision that, for the sake of a better overall lap time, you make a compromise—you drive on a tighter (read slower) arc through the right-hander so that you can position the car at the right side of the road at the entry of the left and get the biggest radius (read faster) arc through the left and onto the straightaway.

This series of corners, the right closely followed by the left, is called a "compromise esse." Any corner where you would make a decision to drive a line different from the optimum big-radius line in order to

accommodate another corner is called a compromise corner. In this first example (Fig. 3-19), the first corner, the right, is completely compromised in order to place the car in position to take the left.

Incomplete Surrender

You don't *always* completely surrender one corner in favor of another. Let's extend the straight between these corners (see Fig. 3-20). Now there's enough room between the turns to let the car come out from the inside edge of the road a quarter of the way and still get back to your turn-in point for the left hander.

By using more road, you've increased the radius of the right, allowing you to take the right a little faster—in this case, at 49 m.p.h. rather than at 47 m.p.h. As the straight between these two corners gets longer, the right has to be compromised to a lesser and lesser extent. At some point you'd be able to run the normal line through the right and still be able to hustle the car over to the turn-in point to the left—the series of corners would no longer be a compromise situation.

The important point here is that some corners are strategically more important than others. We'll discuss this soon in more detail but, before we leave this compromise esse, we should explore other facets of the compromise corner.

Fig. 3-19. Identifying which of two connected corners are more important toward an overall lap time is key in deciding which to compromise.

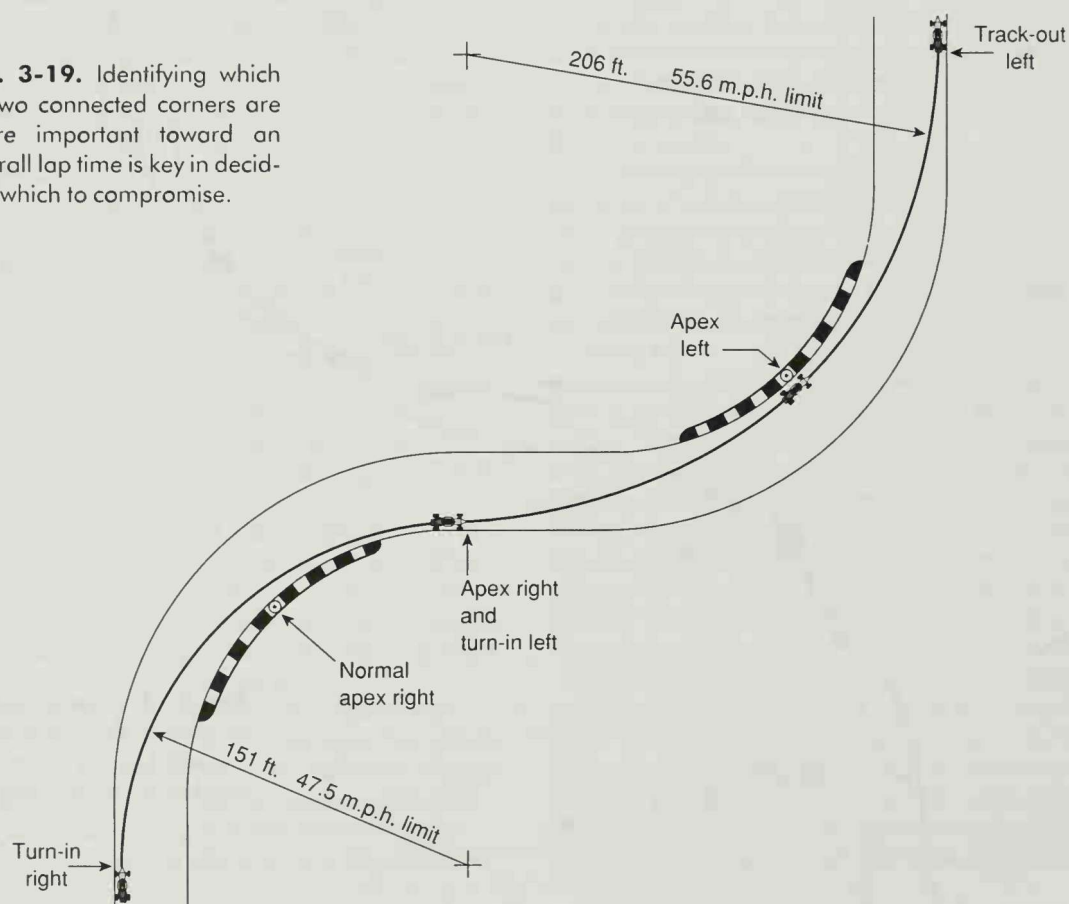
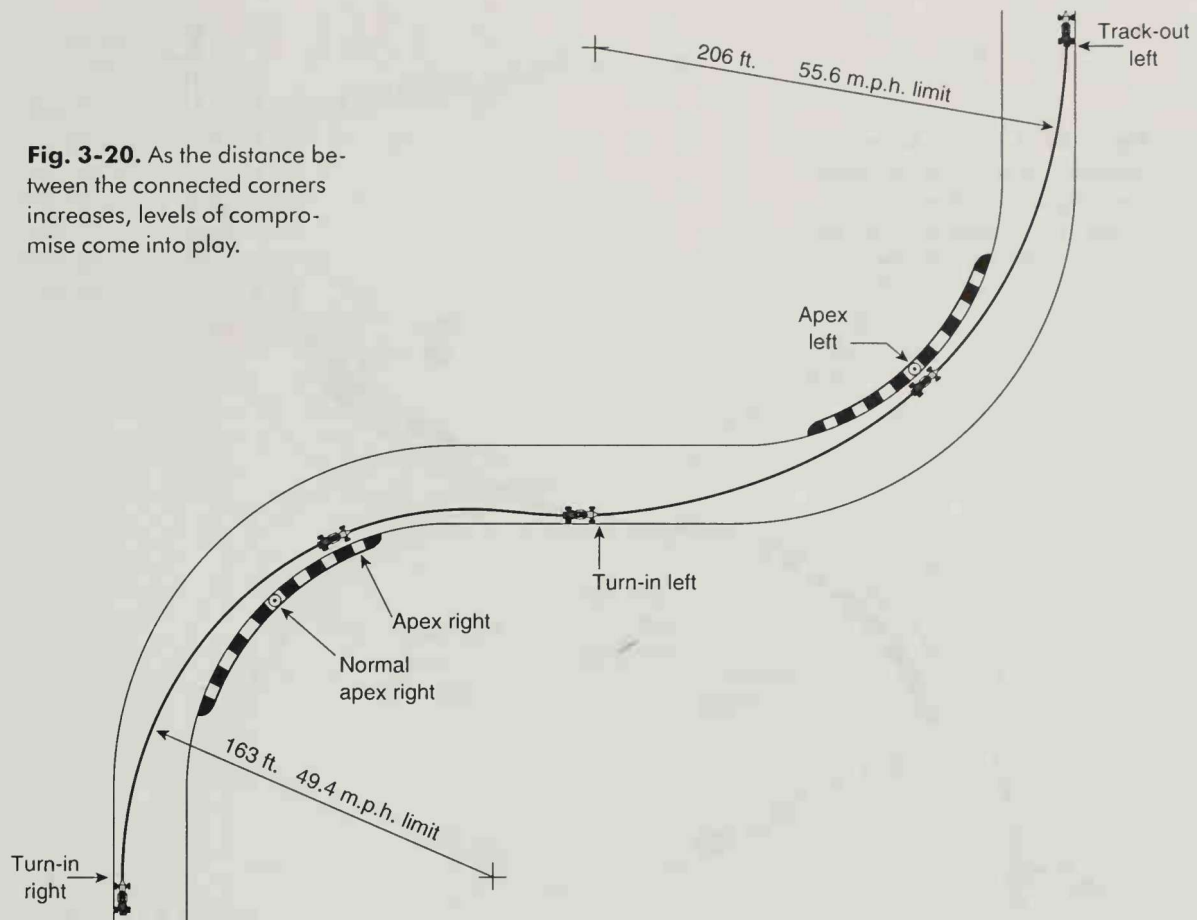


Fig. 3-20. As the distance between the connected corners increases, levels of compromise come into play.



One Line Fits All?

We're often asked whether the line is the line—whether the arcs at low speed in a school race car are the same arcs Indy Car drivers use on the same circuit. The answer is generally yes, with an explanation. All racecars have their strengths and weaknesses. The primary variable, though not the only one, is the ability of the car to accelerate. Let's go back to the compromise *esse* again to see how horsepower can change the line you'd choose.

If you fully compromise the right-hander by taking a very late apex, you would have to run a small radius to get there—in this case, a radius that would have a maximum speed of 47.5 m.p.h. By doing so, as you saw in Fig. 3-19, you make it possible to drive through the left on an arc that would support 55.6 m.p.h. A car with a lot of horsepower will go from 47 m.p.h. to 55 m.p.h. pretty quickly. Herta's line through this *esse* with his Indy Car would be this kind of compromise.

This path wouldn't necessarily be the best for a racecar like a Formula Vee, a single seater powered by a 65 hp four-cylinder Beetle engine. If you slowed to 47 m.p.h. for the right-hander and accelerated as hard as you could through the left the car would only gain, say, 4 m.p.h. because of its limited ability to accelerate

(as in Fig. 3-21). Although the arc through the left would support 55.6 m.p.h., the car can't get there.

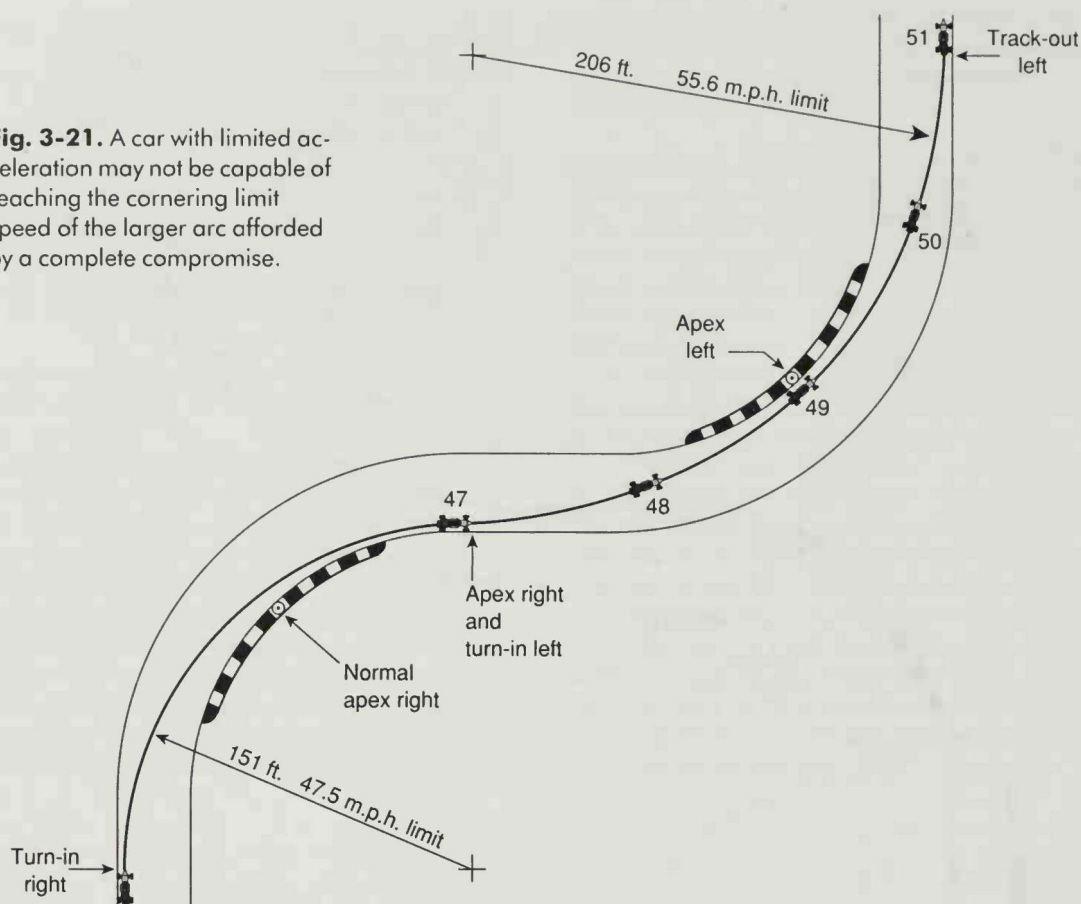
The better compromise for the acceleration-limited car would be to move the apex for the right hand part of the *esse* earlier. This would increase the radius of the right and decrease the radius through the left. This might be a 49 m.p.h. right followed by a 53 m.p.h. left, as in Fig. 3-22. This approach would be 2 m.p.h. faster at the exit of the left than simply appropriating the Indy car line.

GRADING CORNERS

In deciding which corners to compromise, you made a judgment that the corner leading onto the straight was more important than the preceding turn. It's crucial to identify the relative importance of corners on the racetrack.

The first thing to do is to take an overall look at the racetrack and identify the most important corner. Ask yourself where you would most like to have one more mile per hour. One more mile per hour would do you the most good entering the longest period of full acceleration. If you start that acceleration one m.p.h. higher, you'll carry a 1.46 foot per second advantage every second the car is gaining speed.

Fig. 3-21. A car with limited acceleration may not be capable of reaching the cornering limit speed of the larger arc afforded by a complete compromise.



Onto the Longest Straight

When trying to decide which corner to concentrate on most, the corner that leads onto the longest period of effective acceleration is a good starting point. Next is the corner that leads onto the second longest period—next, the third—on and on until you identify the least important corner in terms of exit speed. This grading system is a way to help put your effort in a place where it is most likely to be rewarded.

Fast Corners

There is also a very good argument for putting a high value on fast sweeping corners. The logic involves a number of considerations. The first is that a fast, 90+ m.p.h. turn is a much more intimidating corner to do right on the car's limit. It is much more comfortable sliding through a 50 m.p.h. corner than it is to be on the brink of control at over 90 m.p.h. Consequently, fewer drivers in the field will do the faster corner at its limit. If you are one of the ones that works at generating the confidence it takes to do fast corners at the limit, you'll gain an advantage over the rest.

The second argument is that, since the speed is higher, any percentage loss in potential speed through the turn will represent a higher number. If you can drive at 99% of the car's potential cornering speed,

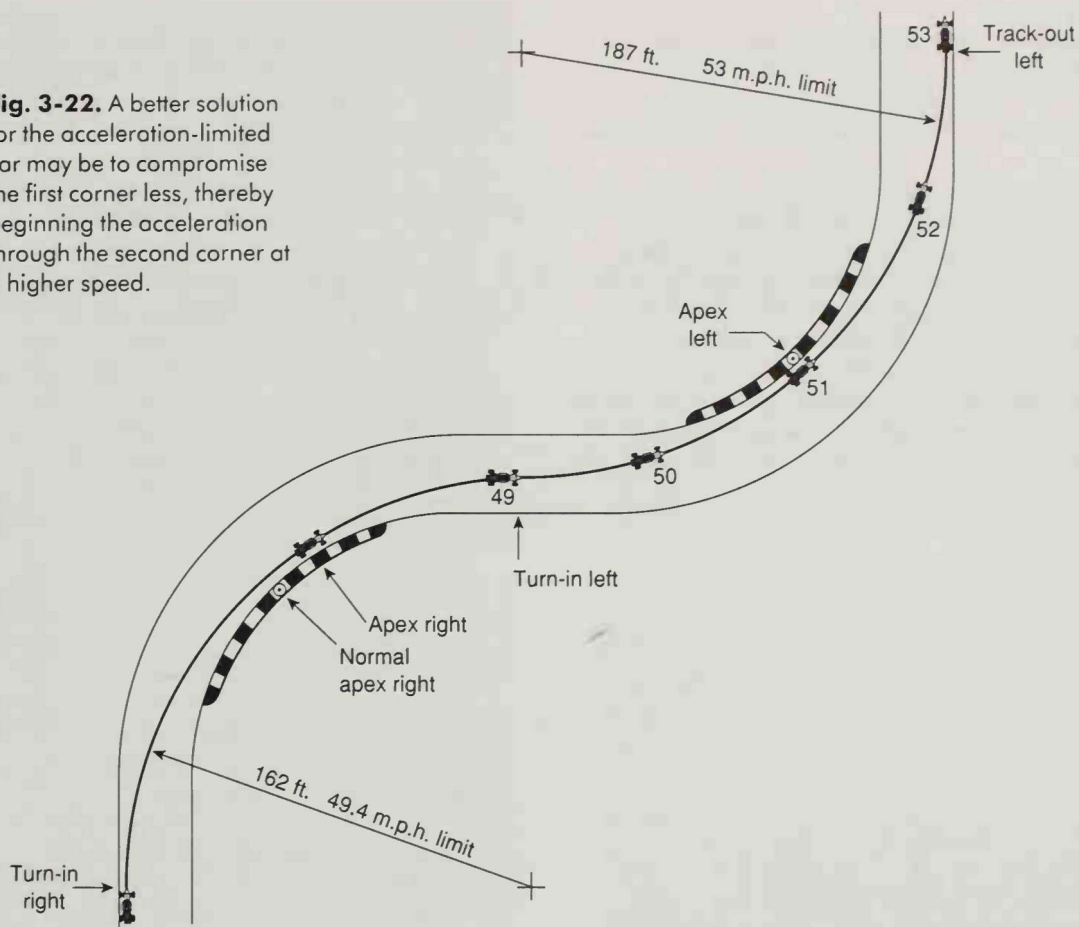
you are .5 m.p.h. below the peak in a 50 m.p.h. corner. In a 90 m.p.h. corner you will be close to a full 1 m.p.h. slower than you could be. Also consider that faster corners are longer than slow corners and if you're right on the limit, carrying that extra 1 m.p.h. over your competitors, you carry that speed over a longer distance than you would in a slower, shorter corner.

Type I, II, Or III

Another system of grading corners is one used by Alan Johnson in his book entitled *Driving in Competition*. Alan points out that there are really three types of turns: turns that lead onto straights, turns at the end of straights, and turns that lead to turns. Turns that lead onto straights he calls Type I turns—turns where exit speed is the most important consideration and therefore would more likely be driven with a later apex to allow for the aggressive acceleration needed to enter the straight at the highest possible speed.

Type II corners are corners at the end of straights and, in order to keep your straightaway speed as long as possible, can frequently be driven with an earlier apex. That way, you can aggressively brake and turn the car down past the apex, not worrying about throttle application and exit speed. If a corner can be a

Fig. 3-22. A better solution for the acceleration-limited car may be to compromise the first corner less, thereby beginning the acceleration through the second corner at a higher speed.



Type I or Type II, it should be treated as a Type I because there is a lot more time spent accelerating from the corner out than decelerating from the braking point in.

The Type III corner is one that leads to another turn. In our example of a compromise esse, the right-hand corner, even though it came at the end of a straightaway, would be considered a Type III, one that needs to be sacrificed for the benefit of a Type I.

KNOWLEDGE IS SPEED

This system of analyzing the corners of the racetrack and making decisions about their relative importance can help prepare you for finding the optimum line *without pushing the starter button of your racecar*. Bill Prout, one of our first instructors and a long-time competitor in a variety of road racing classes, once made the foolish mistake of dividing his annual Formula Atlantic budget by the number of times during the year he pushed the starter button. The resulting dollar-per-start figure was depressingly high, but it's a good reminder that anything you can do to improve

your driving performance without pushing the starter button is valuable indeed.

We've covered many details having to do with the racing line and how to find it. The line, especially if you're fascinated by geometry and mathematics, can be an intriguing subject to play around with. For the sake of clear understanding, we've made some simplifications in the diagrams we've presented. We've drawn constant radius arcs where we've admitted that an absolutely accurate illustration of the car's path through the corner would be on paths with constantly changing radii.

We were tempted to create a computer model that would draw arcs of both decreasing and increasing radii depending on a car's specific braking and accelerating ability and how the driver was using a particular combination of braking and cornering or accelerating and cornering. But for what? In real life you drive a car on a particular path because there are sound logical reasons to choose one route over others. You temper that logic with what the car is telling you as you drive it. What's important is to have the available knowledge and to use it to find a route around the course that takes the least amount of time.

The Key Points

The key points to remember about the line are:

- 1) The path the car takes determines the limit of both your cornering speed and, since it affects your corner exit speed, your ultimate straightaway elapsed time.
- 2) Unless there is a good reason not to, use all the road at the turn-in, apex and track-out.
- 3) There are symptoms for both early and late apexing. Know them and respond to what the car is telling you about the righteousness of your path.
- 4) The best line can change from moment to moment, from day to day, and from car to car.
- 5) The line is seldom a constant radius arc but such an arc is an adequate approximation, especially when you're at the very beginning of the process of finding the line.

"I find that I'm not always right when, ahead of time, I pick what I think are going to be the most important corners—the ones that hold the most potential time in them. Traditional theory has it that the corner leading onto the longest straight is one of the most important. But my experience has found that there are definite exceptions to that. I end up concentrating more on corners leading onto straights initially to find lap time, but as I start driving the circuit I find that certain corners hold more time in them, even though they don't lead onto straights—sometimes a lot more than you would think. Faster corners in general, I find, have the most time in them.

"The way I pick up on it is when we make a change to the racecar and it improves the car in a certain corner more than others and we make a big jump in lap time. I know that the car felt better in only a couple of corners and maybe those corners weren't the ones where I thought there was a lot of time to be gained. Suddenly you realize that there was more time in those corners than you gave them credit for. If I can get better through these corners I'm going to find even more time.

"It's a constant process. Human nature is that you try the hardest in the corners you think you're doing the best in. The corners that you think you're getting through well, you feel comfortable and confident in trying different things and pushing harder. I think it takes a little more of a conscious effort in corners where you're having problems, to look outside of yourself and ask yourself if there's something different you need to do with the car. Is there something different I need to do with my approach to the corner?"

—Bryan Herta